

Patent Keyword Analysis Report

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Detected language: en

Warning: The displayed route is enforced as technical concept → catalog-backed IPC/CPC area → linked FI subdivision → FI-scoped F-term theme/aspect. It is search guidance, not an official classification determination; confirm the current hierarchy and scope in J-PlatPat before relying on it. Each displayed keyword passed the concept-rationale gate: a named technical-concept facet, a server-derived concept basis, and an exact input-text excerpt are required before the result can consume a credit.

AI-derived common technical concept	Shared by all keywords
Object/system	A distributed, online, privacy-preserving machine learning system spanning private data servers and non-private computing devices.
Purpose or problem	Train machine learning models using private local datasets without de-identifying the data or exposing it to unauthorized computing systems.
Application/use	Collaborative training of global machine learning models from knowledge learned across multiple private data holders.
Components	private data servers, local private data, machine learning algorithms, synthetic or proxy data, proxy models, actual trained models, proxy model parameters, non-private computing devices, trained global models
Component relationships	Each private data server trains a machine learning algorithm on its own local private data., Each private data server generates synthetic or proxy data according to the distribution of its actual data., Proxy data is used to train proxy models., Proxy models are compared with trained actual models before learned knowledge is transmitted., Knowledge from multiple private data servers is aggregated into global models without transmitting private data.
Material/composition	—
Manufacturing or processing steps	Generate synthetic or proxy data according to the distribution of actual private data., Train proxy models using the generated proxy data., Assess similarity between proxy models and trained actual models., Transmit proxy data, proxy model parameters, or other learned knowledge when similarity is sufficient., Aggregate learned knowledge from multiple private data servers into trained global models.
Operation	Private servers independently train actual and proxy models, selectively transmit sufficiently representative learned information, and support aggregation into global models.
Control means	Local execution of training on private data servers, Similarity-based transmission decision, Aggregation of learned knowledge rather than private data disclosure
Controlled variable	Model parameters, Proxy-to-actual model similarity, Data distributions, Transmitted learned knowledge
Operating conditions	Private data remains on local private data servers., Transmission occurs when proxy models are sufficiently similar to trained actual models., Private data is not exposed to unauthorized computing systems.
Technical effect	Enables distributed collaborative model training while preserving the confidentiality of local private data and reducing the need for direct data sharing or de-identification.
Context terms	privacy-preserving machine learning, federated learning, distributed training, data confidentiality, model aggregation, synthetic data generation, proxy modeling, local data governance

AI-derived common technical concept

Shared by all keywords

Search phrases

distributed online machine learning using private data servers without exposing local private data, synthetic or proxy data generated according to private data distributions for proxy model training, transmit proxy model parameters when proxy models are sufficiently similar to trained actual models, aggregate learned knowledge from multiple private data servers into a trained global model without private data disclosure

Keyword 1 private data

Normalized term private data

Synonyms confidential data, protected data, sensitive local data

IPC —

CPC —

FI —

F-term —

Occurrences 8

Rank 1

Confidence low

Reason “private data” is grounded in this disclosed relationship between components: “each having local private data” . This specific role makes the term useful for retrieving patents with the same disclosed relationship between components.

Keyword 2 private data servers

Normalized term private data server

Synonyms local data hosts, private data nodes, data-holding servers

IPC —

CPC —

FI —

F-term —

Occurrences 5

Rank 2

Confidence low

Reason “private data servers” is grounded in this disclosed relationship between components: “relevant private data servers train implementations of machine learning algorithms on their local private data” . This specific role makes the term useful for retrieving patents with the same disclosed relationship between components.

Keyword 3	proxy data
Normalized term	proxy data
Synonyms	surrogate data, representative data, distribution-matched data
IPC	—
CPC	—
FI	—
F-term	—
Occurrences	3
Rank	3
Confidence	low
Reason	“proxy data” is grounded in this disclosed processing step: “The private data servers also generate synthetic or proxy data according to the data distributions of the actual data.” . This specific role makes the term useful for retrieving patents with the same disclosed processing step.

Keyword 4	proxy models
Normalized term	proxy model
Synonyms	surrogate models, representative models, stand-in models
IPC	—
CPC	—
FI	—
F-term	—
Occurrences	3
Rank	4
Confidence	low
Reason	“proxy models” is grounded in this disclosed relationship between components: “The servers then use the proxy data to train proxy models.” . This specific role makes the term useful for retrieving patents with the same disclosed relationship between components.

Keyword 5	learned knowledge
Normalized term	learned model knowledge
Synonyms	learned information, training knowledge, model-derived knowledge
IPC	—
CPC	G06T 2207/20081
FI	—
F-term	—
Occurrences	3
Rank	5
Confidence	medium

Keyword 5	
learned knowledge	
Reason	“learned knowledge” is grounded in this disclosed technical operation: “the proxy data, proxy model parameters, or other learned knowledge can be transmitted” . This specific role makes the term useful for retrieving patents with the same disclosed technical operation.
Keyword 6	
machine learning algorithms	
Normalized term	machine learning algorithm
Synonyms	learning algorithms, statistical learning methods, model-training algorithms
IPC	—
CPC	—
FI	—
F-term	—
Occurrences	2
Rank	6
Confidence	low
Reason	“machine learning algorithms” is grounded in this disclosed technical operation: “train implementations of machine learning algorithms on their local private data” . This specific role makes the term useful for retrieving patents with the same disclosed technical operation.
Keyword 7	
proxy model parameters	
Normalized term	proxy model parameter
Synonyms	surrogate-model parameters, proxy weights, learned proxy weights
IPC	—
CPC	—
FI	—
F-term	—
Occurrences	1
Rank	7
Confidence	low
Reason	“proxy model parameters” is grounded in this disclosed controlled variable: “the proxy data, proxy model parameters, or other learned knowledge can be transmitted” . This specific role makes the term useful for retrieving patents with the same disclosed controlled variable.
Keyword 8	
de-identification	
Normalized term	data de-identification
Synonyms	anonymization, identity removal, privacy scrubbing
IPC	—
CPC	—

Keyword 8	de-identification
FI	—
F-term	—
Occurrences	1
Rank	8
Confidence	low
Reason	“de-identification” is grounded in this disclosed technical effect: “without requiring de-identification of the private data” . This specific role makes the term useful for retrieving patents with the same disclosed technical effect.

Search Query Strategies	Database-specific starting points
AI query review - accepted	AI reviewed the generated search-query options, and server validation retained only approved keyword options and database-verified CPC candidates.
Recommended strategy	Query 2 - Balanced. Query 1 is broader; Query 3 is more selective.

Google Patents	Google Patents Advanced Search
Platform note	Uses Boolean keyword groups plus database-verified CPC codes when available.
Query 1 - Broad discovery	High-recall text search for terminology, neighboring concepts, and seed documents. ("private data" OR "confidential data" OR "protected data") AND ("private data server" OR "private data servers" OR "local data hosts")
Query 2 - Balanced (recommended)	Recommended starting point: combines the principal concepts with verified CPC scope when available. ("private data" OR "confidential data" OR "protected data") AND ("private data server" OR "private data servers" OR "local data hosts") AND ("proxy data" OR "surrogate data" OR "representative data") AND CPC=G06T2207/20081
Query 3 - Precision	Uses additional concept groups when available; otherwise narrows synonym alternatives while retaining verified CPC scope. ("private data" OR "confidential data" OR "protected data") AND ("private data server" OR "private data servers" OR "local data hosts") AND ("proxy data" OR "surrogate data" OR "representative data") AND ("proxy model" OR "proxy models" OR "surrogate models") AND ("learned model knowledge" OR "learned knowledge" OR "learned information") AND CPC=G06T2207/20081

PATENTSCOPE	PATENTSCOPE Advanced Search
Platform note	Targets EN_ALLTXT and adds CPC (or IPC fallback) classification syntax when verified codes are available.
Query 1 - Broad discovery	High-recall full-text search using the strongest concept groups. EN_ALLTXT:(("private data" OR "confidential data" OR "protected data") AND ("private data server" OR "private data servers" OR "local data hosts"))
Query 2 - Balanced (recommended)	Recommended starting point: full-text concepts plus a verified classification condition when available. EN_ALLTXT:(("private data" OR "confidential data" OR "protected data") AND ("private data server" OR "private data servers" OR "local data hosts") AND ("proxy data" OR "surrogate data" OR "representative data")) AND CPC:(G06T2207/20081)

PATENTSCOPE		PATENTSCOPE Advanced Search	
Query 3 - Precision		Uses additional concept groups when available; otherwise narrows synonym alternatives while keeping the verified classification condition. EN_ALLTXT:(("private data" OR "confidential data" OR "protected data") AND ("private data server" OR "private data servers" OR "local data hosts") AND ("proxy data" OR "surrogate data" OR "representative data") AND ("proxy model" OR "proxy models" OR "surrogate models") AND ("learned model knowledge" OR "learned knowledge" OR "learned information")) AND CPC:(G06T2207/20081)	
USPTO Patent Public Search		Patent Public Search Advanced Search	
Platform note		Uses Boolean text terms and the .cpc. field code with spaces removed from CPC symbols.	
Query 1 - Broad discovery		High-recall Boolean text search for U.S. patents and published applications. ("private data" OR "confidential data" OR "protected data") AND ("private data server" OR "private data servers" OR "local data hosts")	
Query 2 - Balanced (recommended)		Recommended starting point: principal text concepts plus verified CPC scope when available. ("private data" OR "confidential data" OR "protected data") AND ("private data server" OR "private data servers" OR "local data hosts") AND ("proxy data" OR "surrogate data" OR "representative data") AND G06T2207/20081.cpc.	
Query 3 - Precision		Uses additional concept groups when available; otherwise narrows synonym alternatives while retaining the CPC field restriction. ("private data" OR "confidential data" OR "protected data") AND ("private data server" OR "private data servers" OR "local data hosts") AND ("proxy data" OR "surrogate data" OR "representative data") AND ("proxy model" OR "proxy models" OR "surrogate models") AND ("learned model knowledge" OR "learned knowledge" OR "learned information") AND G06T2207/20081.cpc.	
J-PlatPat		Patent/Utility Model Search - Logical Expression Input	
Platform note		Paste each expression into J-PlatPat's Logical Expression Input. The generator uses * for AND, + for OR, /TX for Full text, verified /FI when available (otherwise /IP for IPC), and /FT for verified F-term refinement.	
Query 1 - Broad discovery		High-recall J-PlatPat full-text logical expression without a classification restriction. [('private data'+ 'confidential data'+ 'protected data')/TX]*[('private data server'+ 'private data servers'+ 'local data hosts')/TX]	
Query 2 - Balanced (recommended)		Recommended starting point: principal full-text concepts combined with verified FI scope, or IPC when verified FI is unavailable. [('private data'+ 'confidential data'+ 'protected data')/TX]*[('private data server'+ 'private data servers'+ 'local data hosts')/TX]*[('proxy data'+ 'surrogate data'+ 'representative data')/TX]	
Query 3 - Precision		Uses additional full-text concept groups when available and adds verified F-term refinement when available. [('private data'+ 'confidential data'+ 'protected data')/TX]*[('private data server'+ 'private data servers'+ 'local data hosts')/TX]*[('proxy data'+ 'surrogate data'+ 'representative data')/TX]*[('proxy model'+ 'proxy models'+ 'surrogate models')/TX]*[('learned model knowledge'+ 'learned knowledge'+ 'learned information')/TX]	

These database-specific queries are search starting points. Verify the current syntax and classification scope in the target system, review the retrieved results, and refine the strategy before relying on it.